

# General Exam Summer 2007

Algebra

August 20, 2007

## **Rules**

1. This is a closed book exam. To use a result, you can cite it by name (e.g., say "by the Main Theorem on Finitely Generated Modules over PIDs") or, if the result does not have a common name, you can just restate it (e.g., say "We know from class that the polynomial ring over a UFD is a UFD").
2. Make sure that I can understand what you write. It will help if you use complete sentences to communicate your ideas. I do not engage in reading minds. You really have to tell me what is going on. Make sure that it is impossible to misunderstand your write-up. I can be somewhat dense, and that will work to your disadvantage.
3. If you think a problem needs clarification, please ask. I will respond to the class.
4. There is a total of 50 points on this exam.
5. This exam has 9 problems and 10 pages. Make sure that none are missing in your copy.

## **Hints**

1. Problems are not sorted according to difficulty.

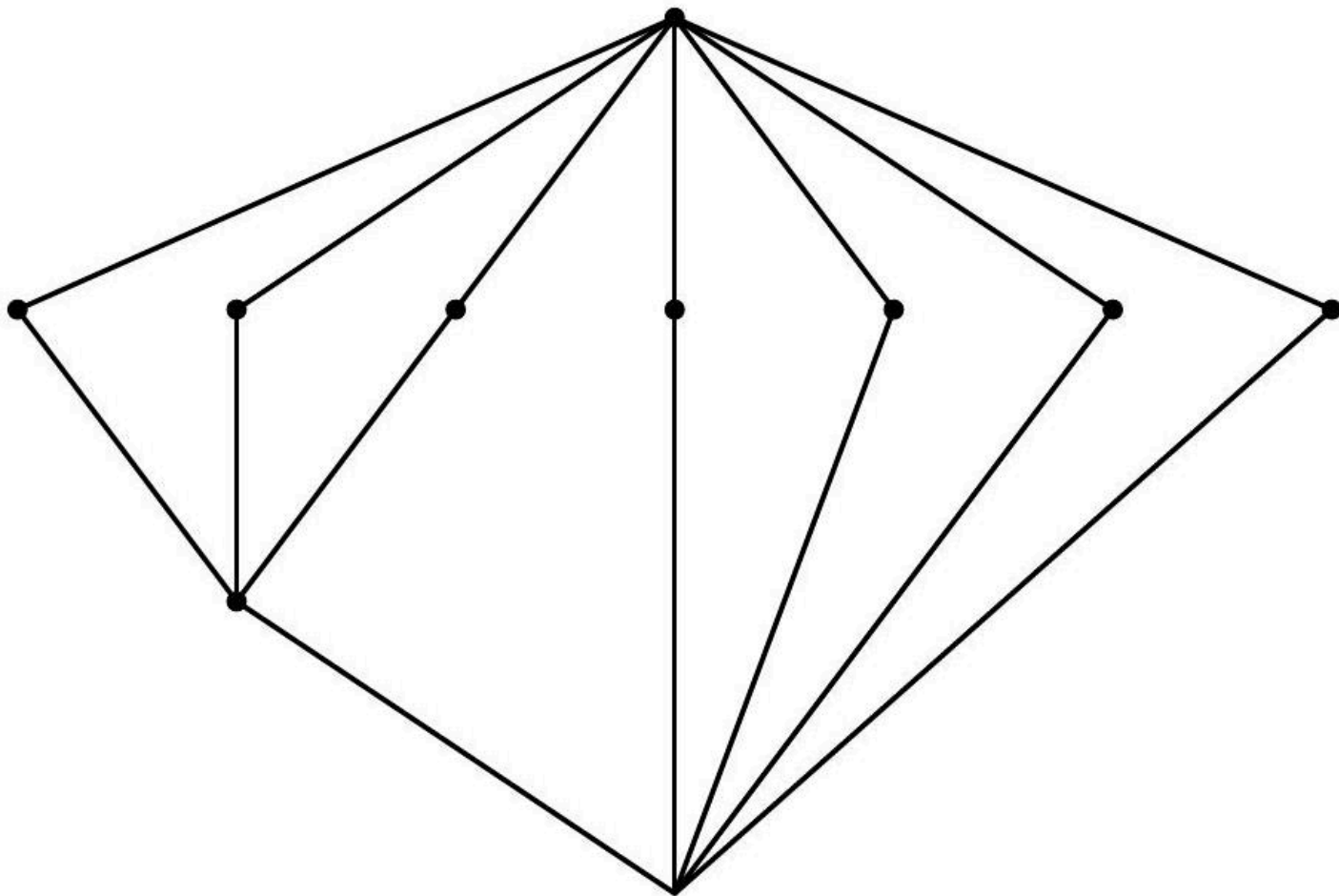
## **Room for your pledge:**

### **Problem 1 [5 points]**

Prove or disprove: the alternating group  $A_{2007}$  has a generating set that consists entirely of elements of order 2007.

### **Problem 2 [5 points]**

Let  $M/K$  be a Galois extension of degree 12. The following is the lattice of intermediate extensions.



The lattice has the top field  $M$ , the bottom field  $K$ , seven vertices on an upper intermediate level, and one vertex on a lower intermediate level. Name the seven upper vertices  $U_1$  through  $U_7$  from left to right, and the lower intermediate vertex  $V$ . Each  $U_i$  is joined upward to  $M$ . The vertices  $U_1$ ,  $U_2$ , and  $U_3$  are joined downward to  $V$ , and  $V$  is joined downward to  $K$ . Each of  $U_4$ ,  $U_5$ ,  $U_6$ , and  $U_7$  is joined directly downward to  $K$ . These are all the edges shown. The auxiliary names  $U_i$  and  $V$  describe positions in the picture; the original vertices are unlabeled.

Here, vertices of the same height may correspond to intermediate extension of different degree. The top vertex corresponds to  $M$  and the bottom vertex corresponds to  $K$ . Show that the Galois group of  $M/K$  is  $A_4$ .

### **Problem 3 [5 points]**

The aim of this problem is to show that finite non-abelian simple groups do not arise as subgroups of  $GL_2(\mathbb{C})$ . You may use the Feit-Thompson theorem that says that finite non-abelian simple groups have even order.

1. Show that there is exactly one element of order 2 in  $SL_2(\mathbb{C})$  and determine that matrix.

2. Show that any finite non-abelian simple subgroup of  $\mathrm{GL}_2(\mathbb{C})$  is contained in  $\mathrm{SL}_2(\mathbb{C})$ .
3. Deduce that  $\mathrm{GL}_2(\mathbb{C})$  does not contain any finite non-abelian simple subgroups.

### **Problem 4 [5 points]**

Let  $M := \mathbb{C}(t)$  be the rational function field, and consider the two  $\mathbb{C}$ -automorphisms  $\varphi$  and  $\psi$  of  $M$  defined by  $\varphi(t) = it$  and  $\psi(t) = t^{-1}$ .

1. Determine the group  $G := \langle \varphi, \psi \rangle \leq \mathrm{Aut}_{\mathbb{C}}(M)$  generated by  $\varphi$  and  $\psi$ .
2. Let  $K := \mathrm{Fix}_M(G)$  be the fixed field of  $G$ . Show that  $G = \mathrm{Aut}_K(M)$  and that  $t^4 + t^{-4}$  generates  $K$  as a field extension over  $\mathbb{C}$ .

### **Problem 5 [5 points]**

Let  $A$  be an  $n \times n$  square matrix with rational coefficients. Suppose that  $A$  has order 5 and assume that the equation  $A\mathbf{v} = \mathbf{v}$  is only satisfied for  $\mathbf{v} = 0$ . Show that  $n$  is divisible by 4.

### **Problem 6 [5 points]**

Show that there is some  $n \in \mathbb{N}$  so that  $\mathbb{Z}[in]$  is not a PID.

### **Problem 7 [5 points]**

Show that

$$p(x) = x^3 - 3x^2 + 15x - 7$$

is irreducible in  $\mathbb{Q}[x]$ .

### **Problem 8 [5 points]**

Let  $M$  be the  $\mathbb{Z}$ -submodule of  $\mathbb{Z}^2$  generated by

$$\begin{pmatrix} 2 \\ 10 \end{pmatrix}, \quad \begin{pmatrix} -2 \\ 8 \end{pmatrix}, \quad \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \quad \begin{pmatrix} -2 \\ 2 \end{pmatrix}$$

Find a basis for  $M$  and determine the structure of  $\mathbb{Z}^2 / M$  as an Abelian group.

## ***Problem 9 [10 points]***

For the following, no reasoning is required:

1. True or false: in a PID, irreducible elements are prime.
2. True or false: in a domain, irreducible elements are prime.
3. True or false: in a PID, prime elements are irreducible.
4. True or false: in a domain, prime elements are irreducible.
5. True or false: a solvable-by-solvable group is solvable.
6. Compute  $(\mathbb{Z}_2 \oplus \mathbb{Q}) \otimes_{\mathbb{Z}} (\mathbb{Z}_2 \oplus \mathbb{Z}_5 \oplus \mathbb{Z})$ .
7. State the definition of a prime ideal.
8. State Hilbert's basis theorem for polynomial rings.
9. State the definition of a solvable group.
10. State the definition of a Euclidean ring.